Mechanics Year 1 · Chapter 11

Variable Acceleration

Functions of time

Differentiation and turning points

Integration and distance travelled

1. A particle has velocity $v = \frac{1}{2}t^3$ m/s. Find its velocity after 4 s, and the time when $v = 108$ m/s.

1

2. Another particle has $v = 2t^2 - 16t + 24$, for $t \geq 0$. Find:

2

a. initial velocity;

b. times at rest;

c. time when $v = 64$;

d. greatest speed on $0 \leq t \leq 5$.

d

6. A particle starts at the origin with velocity $v = 8t - 2t^2$ m/s, $t \geq 0$.

a. When is it next instantaneously at rest?

b. Find its greatest speed on $0 \leq t \leq 4$.

a

b

7. A particle leaves O and later returns to O. Its distance from O is modelled by

$$s = 3t^2 - t^3, \quad 0 \leq t \leq T.$$

The model is valid while $s \geq 0$. Find T and explain.

7

7

8. A particle has velocity
$v = \frac{1}{5}(3t^2 - 10t + 3), \quad t \geq 0.$

a. Find the times when it is at rest.

b. Find its greatest speed on $0 \leq t \leq 3$.

a

b

Using differentiation

$v = \dfrac{dx}{dt}, \qquad a = \dfrac{dv}{dt} = \dfrac{d^2x}{dt^2}.$

A particle has $x = t^4 - 32t + 14$ m. Find:

a. velocity at $t = 3$;

b. time at rest;

c. acceleration at $t = 1.5$.

a

b

c

Test Your Understanding — Luisa · 7

A cat's displacement from a house is

$s = t^3 - \frac{3}{2}t^2 - 36t$ m, $\quad t \geq 0$.

a. Find its velocity at $t = 2$.

b. When is it at rest?

c. Find its acceleration at $t = 5$.

a

b

c

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5. A particle has velocity $v = kt - 3t^2$ m/s. Its initial acceleration is 4 m/s^2.

a. Find k.

b. Find its acceleration when it is next instantaneously at rest.

a

b

6. A printer head has displacement
$s = \frac{1}{4}(4t^3 - 15t^2 + 12t + 30)$ cm,
$0 \leq t \leq 3$.

Find the distance between the points at which it is instantaneously at rest, to 1 d.p.

6

6

A yo-yo's distance below the hand is
$s = 0.6t + 0.4t^2 - 0.2t^3, \quad 0 \leq t \leq 3$.

a. Justify the time restriction.

b. Find the maximum distance from the hand to 3 s.f.

a

b

Test Your Understanding — Luisa · 11

1. A dolphin's velocity is $v = t^3 - 16t^2 + 64t$ m/s until the first maximum, after which it maintains that velocity. Find the time and maximum velocity.
2. A particle has $v = 2t^2 - 14t + 20$, $t \geq 0$. Find its times at rest and greatest speed on $0 \leq t \leq 4$.

1

2

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5. A particle starts at O; its distance from O is
   $s = \frac{9}{2}t^2 - t^3, \quad 0 \leq t \leq 4.5.$

a. Sketch its displacement–time graph.

b. Justify the restriction.

c. Find its greatest distance from O.

d. Find the acceleration magnitude there.

a · Sketch

a–b

c

d

6. A train moves along a straight 4 km test track. Its displacement from the start is modelled by

   $s = 3.6t + 1.76t^2 - 0.02t^3$.

   $0 \le t \le 90$.

Show that it never reaches the far end of the track.

6

6

Using integration

$$v = \int a\,dt, \qquad x = \int v\,dt.$$

Use initial conditions to find each constant.

A particle has $v = 6t - t^2$, $t \ge 0$, and $x(0) = 5$.

a. Find $x(t)$.

b. Find its distance from its starting position at $t = 6$.

a

b

Further Example

A particle has velocity $v = 5 - 3t^2$ m/s, $t \geq 0$. Find the distance travelled in the third second of its motion.

Further Example

A particle has $v = 2t^2 - 9t + 4$ m/s and $x(0) = 15$ m.

a. Find the times at rest.

b. Find acceleration at $t = 5$.

c. Find total distance travelled on $0 \leq t \leq 5$.

a

c

8. A particle starts at O with velocity $v = 6t^2 - 26t + 15$ m/s.

Find the two non-zero times when it passes through O.

8

8

9. A particle starts at O with velocity $v = 3t^2 - 12t + 5$ m/s.

a. When does it return to O?

b. Find the distance travelled on $2 \leq t \leq 3$.

a

b

10. A particle has acceleration $a = 4t - 3$ m/s^2 and initial velocity 4 m/s. At $t = T > 0$, its velocity is again 4 m/s. Find T.

10

10

Constant acceleration formulae

A particle has constant acceleration a, initial velocity u, and initial displacement zero. Use integration to prove:

a. $v = u + at$;

b. $s = ut + \frac{1}{2}at^2$.

a

b

3. Prove that $s = ut + \frac{1}{2}at^2$, with u, a constant, describes constant acceleration a.
4. Which displacement functions have constant acceleration?

A. $s = 2t^2 - t^3$

B. $s = 4t + 7$

C. $s = t^2/4$

D. $s = 3t - 2/t^2$

E. $s = 6$

6. A train passes A at $t = 0$ and B at $t = 40$. Its distance from A is

   $s = 25t - 0.2t^2, \quad 0 \le t \le 40.$

   a. Find AB.

   b. Show acceleration is constant.

   c. At $t = 0$, a bird passes B towards A at 7 m/s, with constant acceleration 0.6 m/s² towards A. How far from A is it when directly above the train?

a

c